

DARSHAN K

Embedded Systems & Firmware Engineer

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PROFESSIONAL SUMMARY

Results-driven Embedded Systems & Firmware Engineer current M.Tech candidate with 2+ years of hands-on experience designing low-level firmware, PCB modules, and hardware-software validation. Expertise in STM32, ESP32, communication protocols (I2C, SPI, UART, USB), and real-time systems.

TECHNICAL SKILLS

Core Programming: Embedded C, C, Python, Shell Scripting (Linux Bash)
MCU & Architectures: ARM Cortex-M (STM32), ESP32 (Xtensa/RISC-V), ESP8266, Raspberry Pi
Hardware Protocols: SPI, I2C, UART, USB, MQTT, AT Commands, CAN Bus
IDE & Toolchains: STM32CubeIDE, Keil uVision5, ESP-IDF, GCC Toolchain
Hardware Design: KiCad (Schematic & PCB), Logic Analyzers, Oscilloscopes, Multimeters
OS / RTOS: FreeRTOS, Linux (Ubuntu), Windows

WORK EXPERIENCE

TechVoc Finesse Consultancy Pvt. Ltd. | Bengaluru

Jan 2025 – Nov 2025

Technical Engineer

- Designed core firmware in Embedded C optimized for low-latency performance on STM32 microcontrollers.
- Engineered custom PCB modules using KiCad, optimizing trace routes for signal integrity.
- Led hardware-software integration debug cycles using logic analyzers and digital oscilloscopes.
- Mentored junior trainees in circuit design, firmware architecture, and PCB validation workflows.

Univision Technology Consulting Pvt. Ltd. | Bengaluru

Aug 2024 – Jan 2025

Graduate Engineering Trainee

- Deployed firmware modules in Embedded C for STM32, ESP32, and Raspberry Pi platforms.
- Configured low-level peripheral drivers for SPI, I2C, and UART; improved data throughput by 12%.
- Executed system-level debugging on Intel architecture prototypes to isolate performance anomalies.
- Streamlined firmware builds via cross-compilation and Linux CLI shell debugging environments.

INTERNSHIP EXPERIENCE

Tescom Technologies Pvt. Ltd.

Nov 2025 – May 2026

IoT, Embedded, and STEM Technology – Intern | Mentor: Mrs. Shruti Patil, Senior Innovation Engineer

- Selected to contribute to IoT and Embedded systems development initiatives under direct mentorship of the Senior Innovation Engineer.
- Working on assigned projects related to IoT, Embedded systems, and STEM technology at Tescom Technologies Office or assigned client locations.

PECSOL TECHNOLOGIES Pvt. Ltd. | India

Nov 2023 – May 2024

Embedded Firmware Intern

- Programmed and validated applications for STM32 and ESP32 boards in STM32CubeIDE and Keil uVision5.
- Contributed to an electric cycle management project, validating sensor data ingestion logic.
- Tested onboard behavior via AT Commands and MQTT over cellular and Wi-Fi networks.

CORE PROJECTS

Server-Based IoT Access Control System

- Engineered an automated asset deployment stack using ESP32, encrypted MQTT protocols, and SPI optimisation—reducing verification latency by 18%.

Real-Time Flight Monitoring & Telemetry Tracker

- Built an embedded telemetry system in Embedded C handling UART sensor streams; custom string-token parser prevented buffer overruns.

EDUCATION

M.Tech in IoT & Sensor Systems – M S Ramaiah Institute of Technology, Bengaluru | CGPA: 8.86

2025 – 2027

B.E. (Electronics) – Maharaja Institute of Technology, Mysore | CGPA: 7.79

2020 – 2024

CERTIFICATIONS

- Advanced Embedded Systems Validation – eTech Prowess (Feb 2025)
- IoT-Network Specialist Professional Certificate – IT-ITes / Skill India (Mar 2026)
- C Development & Programming Foundations – Great Learning Academy (Jun 2023)